



The supply-chain debate

It is quite well known that the number of components in EVs is much lesser than in ICE vehicles. Patwardhan explains, “These can be catered through MSMEs, who adopt digitalisation into their processes.” This can reduce the production time by almost ten times the usual time! Many EV startups are working on diversifying their supply-chain, particularly when it comes to EV powertrains, thus giving a faster edge to EV adoption in India.

When there are 5-6 components that are being supplied and sourced from different vendors, it puts less strain on the supply-chain in the scenario where one supplier fails to deliver a component. There is an option to fall back on another supplier and maintain business continuity.

them all together in the technology stack, and this is like the foundation of a building. It will govern all the links of the supply-chain,” says Deb Mukherjee, Managing Director, Omega Seiki Mobility.

Leveraging digitalisation for product development

“When we talk about digitalisation in product design, process design, manufacturing engineering etc, even in the era of ICE engines these tools used to be there. We did the designs using CAD and simulation tools that were there,” says Mukherjee.

In the context of EV, however, product-specific digitalisation becomes important because it calls for certain specific design capabilities and simulation. Especially in terms of simulation, there are two specific requirements—accuracy of the simulation and reducing product development time. Simulation tools are being leveraged extensively by successful EV giants like Tesla, who have brought down the product lead time significantly with the increased accuracy in simulation.

Simulation for increased efficiency

Digital tools have also helped in the development of the components of EVs—particularly the ones like battery, motor, charging device, and DC-DC converter, where the development time is reduced by one-fourth and the cost is also optimised because the team knows

where it is moving to. Virtual simulation has been a boon to the industry, especially in terms of the number of iterations that can be made during the product development process, while also being cost-efficient.

“With miniaturisation of systems in virtual simulation, you can enhance the scope of the passenger compartment, if you want, since you can miniaturise it and place it anywhere. This is possible in the digital approach. How many models will you build if you are to adopt the traditional approach? It would also be extremely costly,” explains Samir Kumar Mishra, CTO, JBM Group.

The same can be seen with the digital twin approach where the product will be running in real time along with a twin system in the cloud space. Anytime there is a problem, we can test back to the component level. This is also thanks to the introduction of AI, which allows us to predict certain elements without having to make the product go through them. Anytime a problem arises, it can be simulated artificially, pre-programmed, and there is auto-dynamic mode by default where some similar problems can be seen based on the prevailing environment. That gives enough hints of the problems that could occur.

M.A. Patwardhan, GM-Engineering Design & Simulation, ARAI, notes, “My opinion is that EVs are a better candidate for the digital factories compared to the

traditional factories since Industry 4.0 and other kinds of tools can be employed for a digital factory.”

The main purpose of the digital factory is to validate and optimise the design of different production plans in virtual form during the planning phase itself. This can be seamlessly done through the integration of 3D simulation tools and visualisation tools, radically transforming the industry and production value chain for EVs.

Product design

Diving a little deeper on the design side, the introduction of digitalisation into the product lifecycle management (PLM), where it is equipped with the latest simulation tools, has been a boon. In traditional architecture of the gasoline vehicle, the vehicle is designed around one or two components that are central to the product but become a constraint. When someone wants to replace those components, it becomes difficult to go back to the design and change them. It takes a complete iteration process to do that.

With the digitisation in PLM, it becomes a plug-and-play kind of system where we can just change a couple of components and work around the design right from the beginning with five to six components. Thus, there is not much of a risk in the supply-chain when there is a constraint to a component around which the vehicle production is running. The things are already fitted and plugged and played, and any additional components can be included in the design. This is the power of the PLM.

Another aspect in designing, which has been carried over from the traditional approach, is called ‘Class of Titans’ where two titans—the car designer or stylist and the engineering team—do not see eye to eye at any time. There is always a clash in the traditional